



Báo cáo kho dữ liệu - data warehouse

VKU_Kho dữ liệu (Trường Đại học Công nghệ Thông tin và Truyền thông Việt - Hàn)



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**VIETNAM-KOREA UNIVERSITY OF INFORMATION AND COMMUNICATION
TECHNOLOGY**

COMPUTER SCIENCE DEPARTMENT



DATA WAREHOUSE

**BUILDING DATA WAREHOUSE FOR ORDER
DELIVERY AND MANAGEMENT**

Students : **NGUYỄN VĂN DŨNG**

MAI QUỐC HUY

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Instructor: **MD. TRẦN THANH LIÊM**

Da Nang, December 2023

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ACKNOWLEDGEMENT

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During the course of my research project, I have made efforts and worked hard, but I understand that there may still be some shortcomings. I hope to receive valuable feedback from my professors and the evaluation committee to further improve and refine my research work.

I sincerely thank you!

Da Nang, December 2023

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LIST OF ABBREVIATIONS

ID	Phrase	Abbreviation
1	SSMS	SQL Server Management Studio
2	SSIS	SQL Server Integration Services
3	SSRS	SQL Server Reporting Services
4	API	Application Programming Interface
5	GUI	Graphical User Interface
6	IDE	Integrated Development Environment
7	UI	User Interface
8	SQL	Structured Query Language
9	ETL	Extract, Transform, Load

CHAPTER 1: INTRODUCTION

1.1 Project Introduction:

1.1.1 Reasons for choosing the project:

Building a data warehouse for order delivery and management presents a compelling opportunity for businesses to revolutionize their operational efficiency, enhance customer satisfaction, and gain a significant competitive advantage. By consolidating and analyzing data from diverse sources, this project unlocks valuable insights that can transform how businesses manage and optimize their order delivery processes.

Enhanced Visibility and Decision Making: A centralized data warehouse provides a holistic view of all order delivery data, enabling stakeholders to identify trends, patterns, and inefficiencies. This data-driven insight empowers informed decision-making about resource allocation, route optimization, and service improvements, leading to a more efficient and effective order delivery operation.

Improved Customer Satisfaction: Proactive issue resolution and personalized delivery experiences are key to customer satisfaction. A data warehouse facilitates accurate tracking of individual orders and allows businesses to identify potential delays or issues before they impact customers. Additionally, analyzing delivery data reveals customer preferences, enabling businesses to tailor delivery experiences and exceed customer expectations, fostering loyalty and positive brand perception.

Increased Operational Efficiency: Optimizing delivery routes, schedules, and resource allocation is crucial for maximizing operational efficiency. A data warehouse provides valuable insights to identify bottlenecks and inefficiencies in the order delivery process. This enables businesses to streamline operations, reduce costs, and improve overall delivery performance.

Improved Inventory Management: Accurate forecasting and efficient planning are essential for effective inventory management. A data warehouse provides valuable insights into inventory levels and customer demand, enabling businesses to optimize procurement, warehousing, and distribution activities. This reduces stockouts and overstocks, leading to cost savings and increased operational efficiency.

Enhanced Competitive Advantage: Data-driven decision-making and innovative solutions are key to gaining a competitive edge in today's market. A data warehouse empowers businesses to leverage data insights to develop new and improved delivery methods, optimize existing processes, and enhance the overall customer experience. This differentiated approach sets businesses apart from the competition, securing a strong position in the market.

These compelling reasons highlight the significant benefits of building a data warehouse for order delivery and management. By embarking on this project, businesses can unlock a wealth of valuable data insights, transform their order delivery processes, and achieve significant improvements in efficiency, customer satisfaction, and overall competitiveness.

1.1.2 The goals of the project:

The primary purpose of the project "Building a Data Warehouse for Order Delivery and Management" is to centralize, consolidate, and analyze order

delivery data from various sources to gain valuable insights and optimize the entire delivery process. This project aims to transform how businesses manage and deliver orders, leading to a more efficient, effective, and customer-centric operation.

1.2 Tool Introduction:

1.2.1 Visual Studio

Visual Studio is a powerful Integrated Development Environment (IDE) created by Microsoft for building various types of applications. It provides a comprehensive set of tools and features that support the entire development lifecycle, from coding and debugging to testing and deployment.

1.2.1.1 What is Visual Studio?



Visual Studio is an IDE that provides a single platform for developers to create various types of applications, including:

- **Desktop applications:** Develop applications with a graphical user interface (GUI) for Windows.
- **Web applications:** Build web applications and services using technologies like ASP.NET, Node.js, and ASP.NET Core.
- **Mobile applications:** Create native mobile applications for Android, iOS, and Windows using tools like Xamarin and React Native.
- **Cloud applications:** Build and deploy applications and services to Azure, AWS, and other cloud platforms.
- **Games:** Develop games using Unity, Unreal Engine, and other game development tools.
- **Data science and machine learning:** Visual Studio provides tools and libraries for data analysis, machine learning, and artificial intelligence.

1.2.1.2 Key features of Visual Studio

Here are some of the key features that make Visual Studio a popular choice for developers:

- **Code editor:** Visual Studio provides a powerful code editor with syntax highlighting, code completion, and code refactoring features.
- **Debugger:** The debugger allows you to step through your code line by line, set breakpoints, and examine variables to identify and fix errors.
- **Testing tools:** Visual Studio integrates with various testing frameworks like NUnit, MSTest, and xUnit to help you write and run unit tests, integration tests, and UI tests.
- **Version control:** Visual Studio supports popular version control systems like Git and Mercurial to help you manage your code history and collaborate with other developers.

- **Extensibility:** Visual Studio is highly extensible, allowing you to add new features and functionality through extensions.
- **Support for multiple languages:** Visual Studio supports a wide range of programming languages, including C#, C++, VB.NET, F#, Python, JavaScript, and more.

1.2.1.3 Benefits of using Visual Studio

There are many benefits to using Visual Studio for your development projects, including:

- **Increased productivity:** The various features and tools in Visual Studio can help you write code faster and more efficiently.
- **Improved quality:** Visual Studio helps you identify and fix errors early in the development process, resulting in higher quality applications.
- **Collaboration:** Visual Studio provides features that make it easier to collaborate with other developers on your projects.
- **Scalability:** Visual Studio can be used to develop small, simple applications as well as large, complex applications.
- **Customization:** Visual Studio can be customized to meet your specific workflow and preferences.

1.2.2 SSMS

1.2.2.1 What is SSMS ?



SQL Server Management Studio (SSMS) is a powerful and versatile graphical user interface (GUI) tool used to manage and administer Microsoft SQL Server databases. It provides a comprehensive set of features for working with databases, including:

- Connecting to and managing SQL Server instances
- Writing and executing Transact-SQL (T-SQL) queries
- Creating and managing database objects (tables, views, stored procedures, etc.)
- Designing and modifying database schemas
- Administering user permissions and security
- Monitoring and optimizing database performance

SSMS is a crucial tool for anyone who works with SQL Server databases, from developers and database administrators to business analysts and data scientists.

1.2.2.2 Features of SSMS

Here are some of the key features of **SSMS**:

- **Object Explorer:** A hierarchical view of all the objects in a SQL Server instance.
- **Query Editor:** A powerful editor for writing and executing T-SQL queries.
- **Table Designer:** A graphical tool for designing and modifying database tables.
- **Data Viewer:** A grid-based interface for viewing and editing data in tables.
- **Index Tuning Wizard:** A tool for assisting with the creation and optimization of indexes.
- **Backup and Restore Wizard:** A tool for backing up and restoring databases.
- **Security Management:** Tools for managing user permissions and roles.
- **Performance Dashboard:** A graphical interface for monitoring database performance.

1.2.2.3 Benefits of using SSMS

There are many benefits to using **SSMS** for your database administration tasks, including:

- **Increased productivity:** The graphical interface and intuitive features of SSMS make it easy to manage and administer databases.
- **Improved efficiency:** SSMS automates many tasks, such as creating scripts and managing permissions, saving you time and effort.
- **Reduced errors:** SSMS helps to prevent errors by providing features such as syntax highlighting and IntelliSense.
- **Enhanced security:** SSMS provides tools for managing user permissions and roles, helping to keep your databases secure.

- **Improved performance:** SSMS provides tools for monitoring and optimizing database performance, ensuring that your databases run smoothly.

1.2.3 SSIS

1.2.3.1 What is SSIS ?



SQL Server Integration Services (SSIS), a powerful tool within the Microsoft SQL Server suite, transcends the limitations of a mere data extraction, transformation, and loading (ETL) tool. It's a complete platform, an ecosystem, empowering developers and data professionals to build robust, scalable, and maintainable data integration solutions.

1.2.3.2 Features of SSIS

SSIS offers a vast array of features and functionalities, enabling you to:

- **Extract data from diverse sources:** Connect to various data sources, including relational databases, flat files, APIs, and cloud services, for comprehensive data acquisition.
- **Transform and cleanse data:** Apply transformations to restructure, clean, and enrich your data, ensuring its accuracy and consistency for further analysis.

- **Load data into target destinations:** Load your transformed data into a multitude of target destinations, including relational databases, data warehouses, cloud storage, and data lakes.
- **Orchestrate complex data flows:** Build intricate data pipelines with control flow tasks and data flow components, automating data workflows and ensuring data integrity.
- **Monitor and manage your data pipelines:** Leverage the SSIS Management Studio for comprehensive monitoring, logging, and troubleshooting, guaranteeing smooth data integration processes.
- **Extend functionality with custom components:** Develop custom components to address unique integration needs and extend the capabilities of SSIS beyond its core functionality.
- **Leverage the power of the cloud:** Integrate with Azure services for scalable data processing and leverage cloud-based data storage solutions for cost-effective and efficient data management.

1.2.3.3 Benefits of Choosing SSIS

Embrace a world of advantages when you choose **SSIS**:

- **Increased Efficiency:** Automate data integration tasks, eliminating manual work and boosting productivity.
- **Improved Data Quality:** Cleanse and transform data, ensuring its accuracy and consistency for reliable analysis.
- **Scalability and Flexibility:** Design data pipelines to handle large volumes of data and adapt to changing needs.
- **Reduced Costs:** Optimize data processing and leverage cloud services for cost-effective solutions.
- **Enhanced Security:** Integrate with existing security infrastructure and ensure data privacy.

- **Ease of Use:** Utilize a user-friendly interface and visual tools for intuitive data integration design.

1.2.4 SSRS

1.2.4.1 What is SSRS ?



SQL Server Reporting Services (SSRS), crafted by Microsoft, transcends the limitations of a mere reporting tool. It's a robust platform, a complete ecosystem, empowering developers and business users to create and manage comprehensive and insightful reports.

1.2.4.2 Features of SSRS

SSRS offers a vast array of features and functionalities, enabling you to:

- **Design and build interactive reports:** Create visually appealing and interactive reports with tables, charts, graphs, and other visualizations for effective data presentation.
- **Connect to diverse data sources:** Access data from various sources, including relational databases, flat files, APIs, and cloud services, for consolidated reporting.

- **Develop complex reports with rich features:** Utilize powerful features like parameters, filters, and drill-down capabilities to allow users to explore data in-depth and uncover hidden insights.
- **Deliver reports in various formats:** Export reports in multiple formats, including PDF, Word, Excel, and HTML, for convenient distribution and access.
- **Schedule and automate report delivery:** Schedule reports to run automatically at specific times or intervals, ensuring timely and consistent delivery to stakeholders.
- **Manage and secure your reports:** Utilize the web-based Report Manager for centralized report management, access control, and security configuration.
- **Extend functionality with custom extensions:** Develop custom extensions to address unique reporting needs and extend the capabilities of SSRS beyond its core functionality.
- **Leverage the power of the cloud:** Integrate with Azure services for scalable reporting and leverage cloud-based storage solutions for cost-effective and efficient report management.

1.2.4.3 Benefits of Choosing SSRS

SSRS offers a vast array of features and functionalities, enabling you to:

- **Design and build interactive reports:** Create visually appealing and interactive reports with tables, charts, graphs, and other visualizations for effective data presentation.
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CHAPTER 2: ANALYSIS AND DESIGN OF DATA WAREHOUSE

2.1 Logical Modeling

2.1.1 Dimensions

DIM_Time

<i>DIM_Time</i>
timeID
date
year
quater
day

Hình 2.1.1.1 DIM_Time

DIM_Customer

<i>DIM_Customer</i>
Customer_Id
Customer_Fname
Customer_Lname
Customer_Segment
LocationID

Hình 2.1.1.2 DIM_Customer

DIM_Location

DIM_Locaton
locationID
country
city
state
Street

Hình 2.1.1.3 DIM_Location

DIM_Product

DIM_Product
Product_Card_Id
Product_Name
Product_Price

Hình 2.1.1.4 DIM_Product

DIM_Category

DIM_Category
Category_Id
Category_Name

Hình 2.1.1.5 DIM_Category

2.1.2 Facts

FACT_Order

FACT_Order
Type
Order_Id
Order_Item_Discount
Order_Item_Quantity
Sales
Order_Item_Total
Order_Item_Id
TimeID
Product_Card_Id
category_Id
customer_Id
location_Id

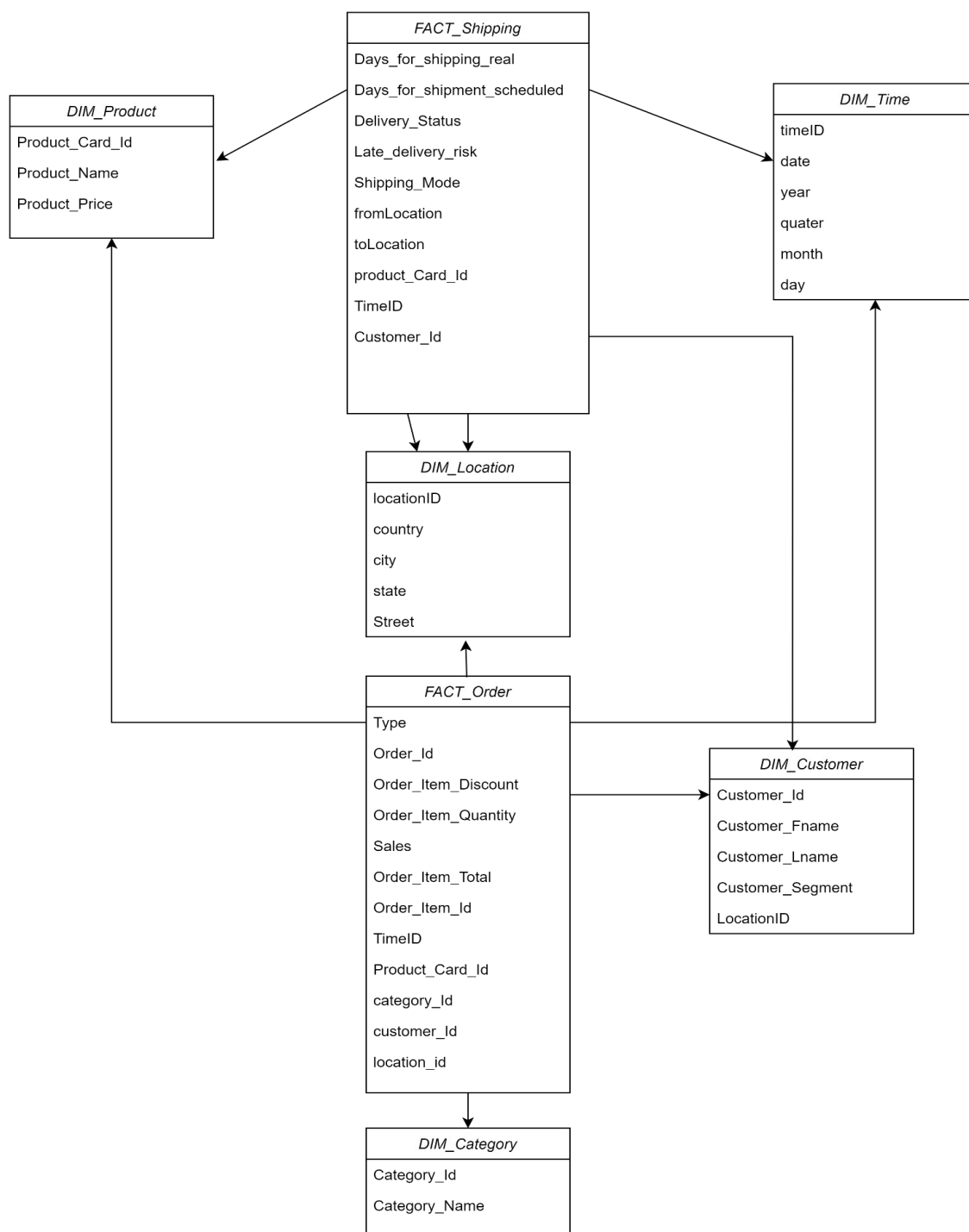
Hình 2.1.2.1 FACT_Order

FACT_Shipping

FACT_Shipping
Days_for_shipping_real
Days_for_shipment_scheduled
Delivery_Status
Late_delivery_risk
Shipping_Mode
fromLocation
toLocation
product_Card_Id
TimeID
Customer_Id

Hinh 2.1.2.2 FACT_Shipping

2.1.3 Constellation Schema



Hình 2.1.3.1 Constellation Schema

2.1 Physical Modeling

2.2.1 Dimensions

Name	Type	Constraint	Comment
DIM_Customer			Customer Dimension table including user's data
Customer_Id	smallint	PRIMARY KEY	Unique ID of a user
Customer_Fname	nvarchar(50)		Customer's First name
Customer_Lname	nvarchar(50)		Customer's Last name
Customer_Country	nvarchar(50)		Customer's Country
Customer_State	nvarchar(50)		Customer's State
Customer_CIty	nvarchar(50)		Customer's City
Customer_Street	nvarchar(50)		Customer's Street
Customer_Segment	nvarchar(50)		Customer's Segment
DIM_Product			Product Dimension table including product's data
Product_Card_Id	smallint	PRIMARY KEY	Product card unique ID
Product_Name	nvarchar(50)		Name of the product

Product_Price	float		Price of the product
DIM_Time			Time Dimension table including data of order timestamps
timeID	numeric(18,0)	PRIMARY KEY	The Unique ID for timestamp
date	datetime2(7)		The exact date time of the timestamps
year	int		The year of the timestamps
quater	int		The quarter of the timestamps
month	int		The month of the timestamps
day	int		The day of the timestamps
DIM_Location			Location Dimension table including the data of a location
locationID	varchar(50)	PRIMARY KEY	Unique ID of a Location
country	nvarchar(50)		Country name of the Location
state	nvarchar(50)		State name of the Location

city	nvarchar(50)		City name of the Location
region	nvarchar(150)		Region name of the Location
DIM_Category			Category Dimension table including data of the Category
Category_Id	tinyint	PRIMARY KEY	Unique ID of the Category
Category_Name	nvarchar(50)		The name of the Category

Bảng 2.2.1.1 Dimensions

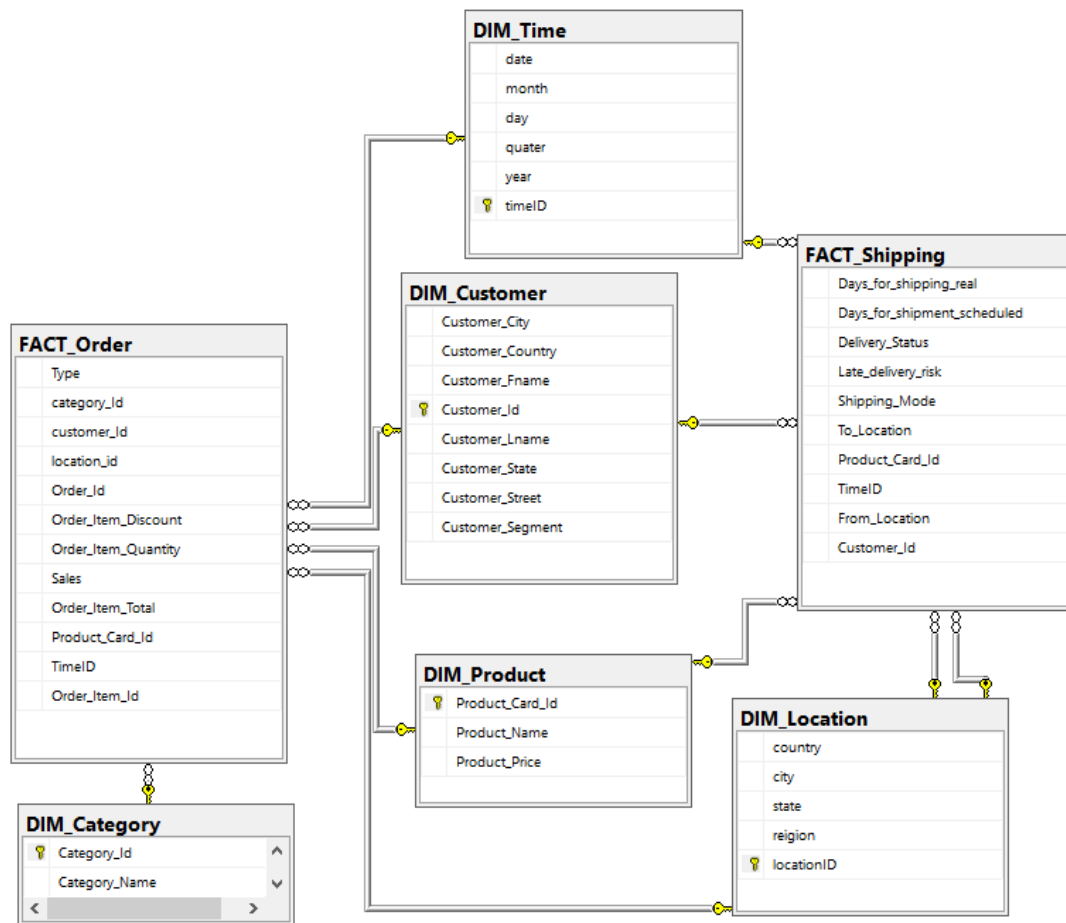
2.2.2 Facts

Name	Type	Comment
FACT_Order		The FACT table containing all the metrics for Orders
category_Id	tinyint	Category ID of the item in the order
customer_Id	smallint	Customer ID of the order's owner
Type	nvarchar(50)	The type of payment method for this order
location_id	varchar(50)	The location ID of the customer
Order_Id	int	The ID for the order
Order_Item_Discount	float	The discount percentage for items of the order
Order_Item_Quantity	tinyint	The amount of items in the order

Sales	float	The cost before discount
Order_Item_TOTal	float	The cost after discount
Product_Card_Id	smallint	The ID of the Product card
TimeID	numeric(18,0)	The ID of the timestamps of the Order
Order_Item_Id	int	The ID of the items in the order
FACT_Shipping		The FACT table containing all the metrics for the Shipping
Days_for_shipping_real	tinyint	The days that delivery actually took
Days_for_shipping_schedule	tinyint	The days for delivery that the system estimate
Delivery_Status	nvarchar(50)	The status of the delivery
Late_delivery_risk	bit	The risk of late delivery estimate by system
Shipping_Mode	nvarchar(50)	The method of Shipping that user selected
To_Location	varchar(50)	The location that the order is delivered to
Product_Card_Id	smallint	The ID of the product card
TimeID	numeric(18,0)	The ID of the timestamps of the order
From_Location	varchar(50)	The location that the order is taken
Customer_Id	smallint	The order's buyer customer ID

Bảng 2.2.2.1 Facts

2.2.3 Physical Model

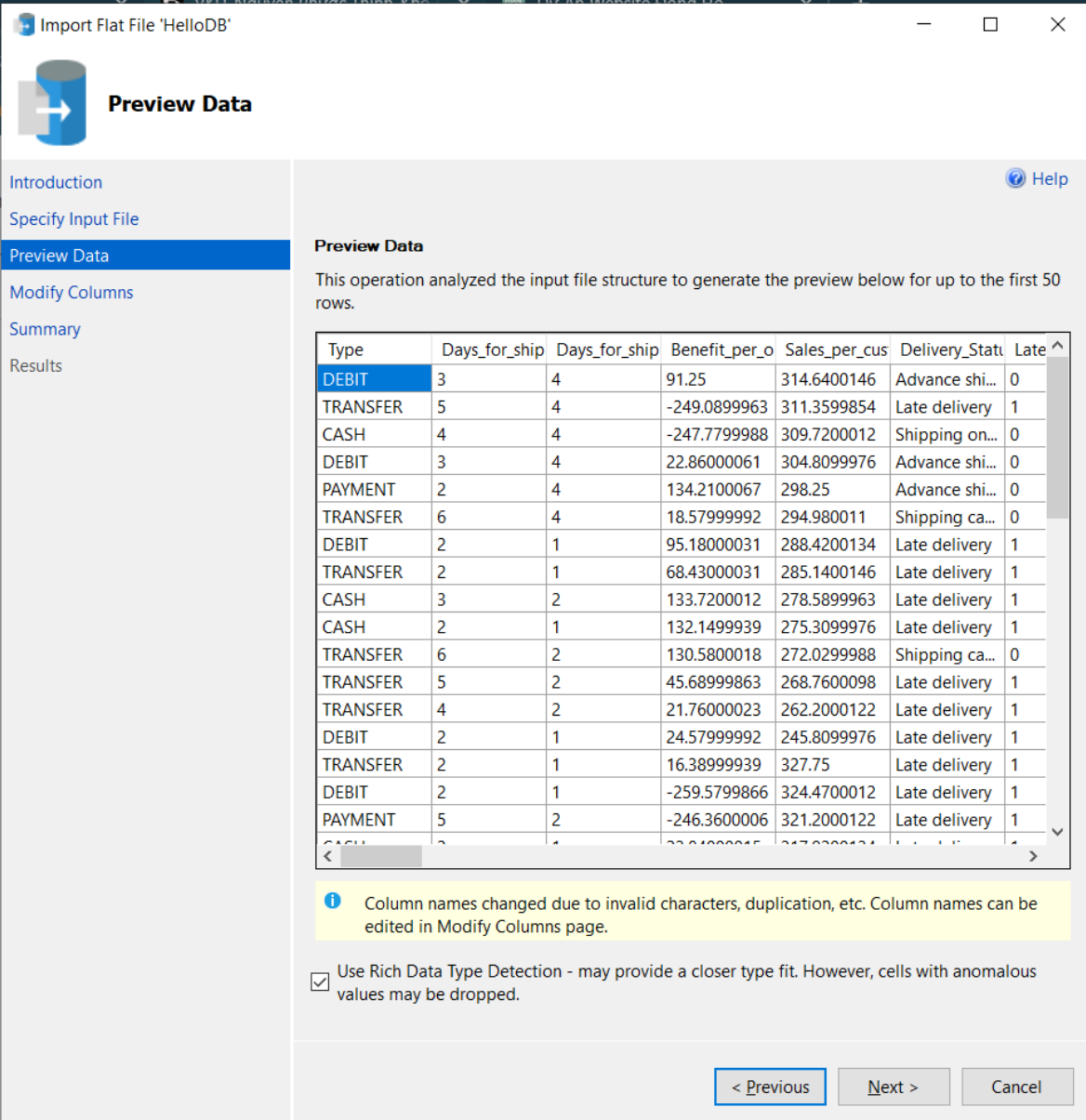


Hinh 2.2.3.1 Physical Model

CHAPTER 3: ETL PROCESS

3.1 Prepare data

Import flat file to Sql Server



The screenshot shows the 'Import Flat File 'HelloDB'' window in SQL Server Data Tools. The 'Preview Data' tab is selected, showing a preview of the data from the flat file. The data is organized into columns: Type, Days_for_ship, Days_for_ship, Benefit_per_o, Sales_per_cus, Delivery_Statu, and Late. The preview shows 20 rows of data, including transactions like DEBIT, TRANSFER, CASH, and PAYMENT. A message at the bottom indicates that column names were changed due to invalid characters or duplication. There are navigation buttons at the bottom: '< Previous', 'Next >', and 'Cancel'.

Preview Data

This operation analyzed the input file structure to generate the preview below for up to the first 50 rows.

Type	Days_for_ship	Days_for_ship	Benefit_per_o	Sales_per_cus	Delivery_Statu	Late
DEBIT	3	4	91.25	314.6400146	Advance shi...	0
TRANSFER	5	4	-249.0899963	311.3599854	Late delivery	1
CASH	4	4	-247.7799988	309.7200012	Shipping on...	0
DEBIT	3	4	22.86000061	304.8099976	Advance shi...	0
PAYMENT	2	4	134.2100067	298.25	Advance shi...	0
TRANSFER	6	4	18.57999992	294.980011	Shipping ca...	0
DEBIT	2	1	95.18000031	288.4200134	Late delivery	1
TRANSFER	2	1	68.43000031	285.1400146	Late delivery	1
CASH	3	2	133.7200012	278.5899963	Late delivery	1
CASH	2	1	132.1499939	275.3099976	Late delivery	1
TRANSFER	6	2	130.5800018	272.0299988	Shipping ca...	0
TRANSFER	5	2	45.68999863	268.7600098	Late delivery	1
TRANSFER	4	2	21.76000023	262.2000122	Late delivery	1
DEBIT	2	1	24.57999992	245.8099976	Late delivery	1
TRANSFER	2	1	16.38999939	327.75	Late delivery	1
DEBIT	2	1	-259.5799866	324.4700012	Late delivery	1
PAYMENT	5	2	-246.3600006	321.2000122	Late delivery	1
CASH	3	2	133.7200012	278.5899963	Late delivery	1

Column names changed due to invalid characters, duplication, etc. Column names can be edited in Modify Columns page.

☒ Use Rich Data Type Detection - may provide a closer type fit. However, cells with anomalous values may be dropped.

< Previous Next > Cancel

Figure 3.1 Import flat file to sql server

3.2 Create new project ssis

Select Integration Services Project

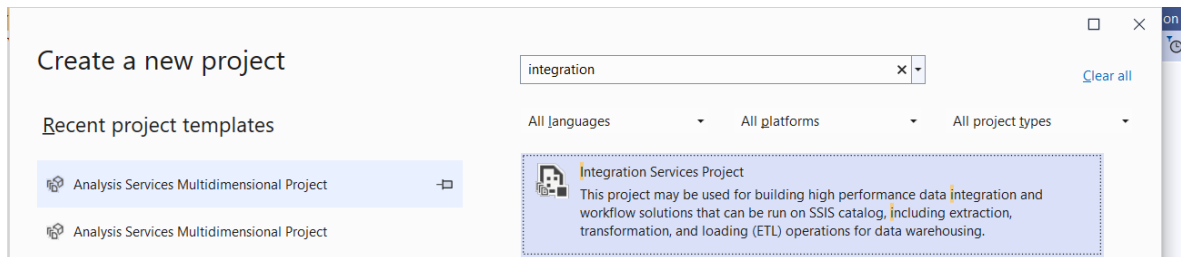


Figure 3.2.1 Select Integration Services Project

Configure your new project

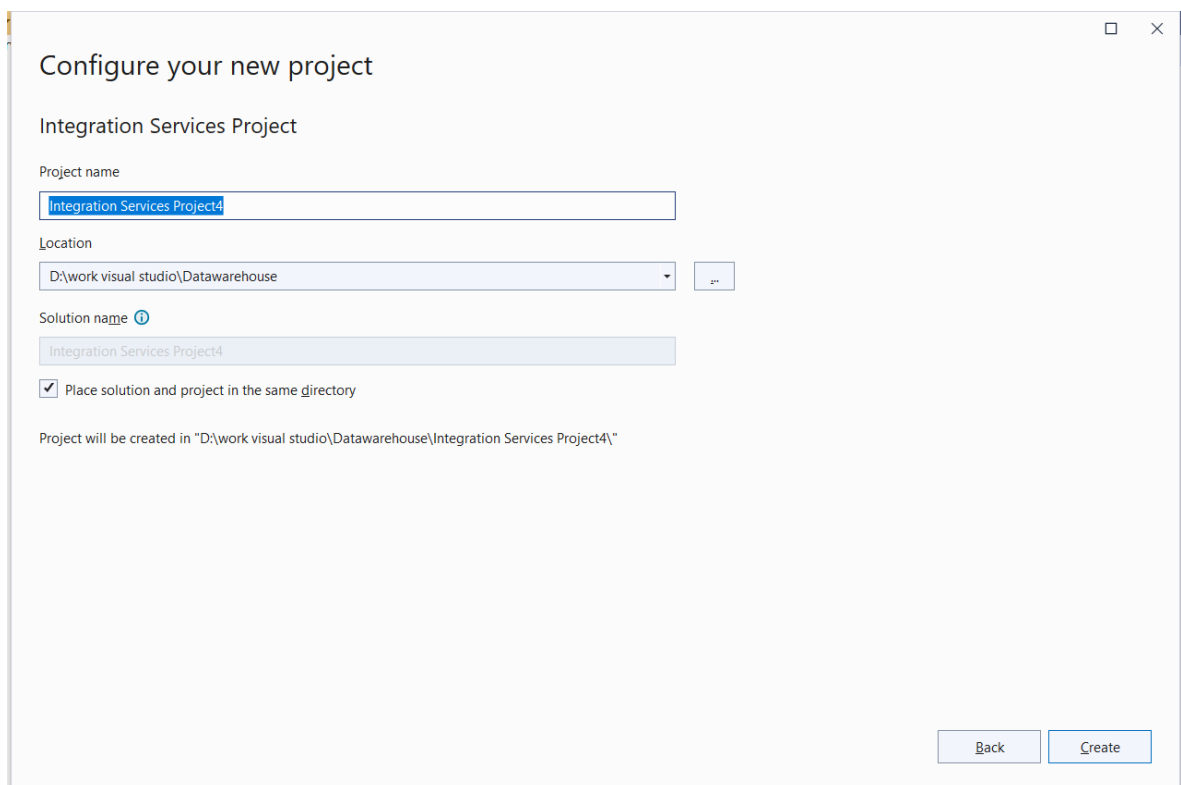


Figure 3.2.2 Configure your new project

Project after creation

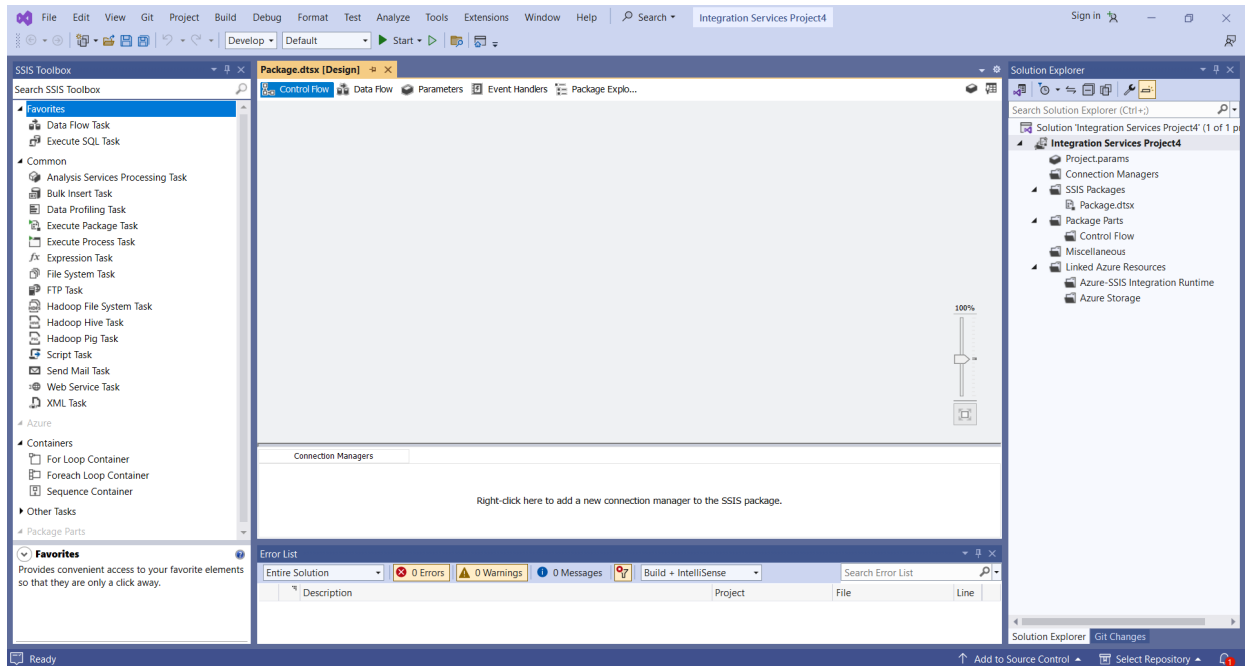


Figure 3.2.3 Project after creation

3.3 ETL

3.3.1 New Source assistant from Sql Server

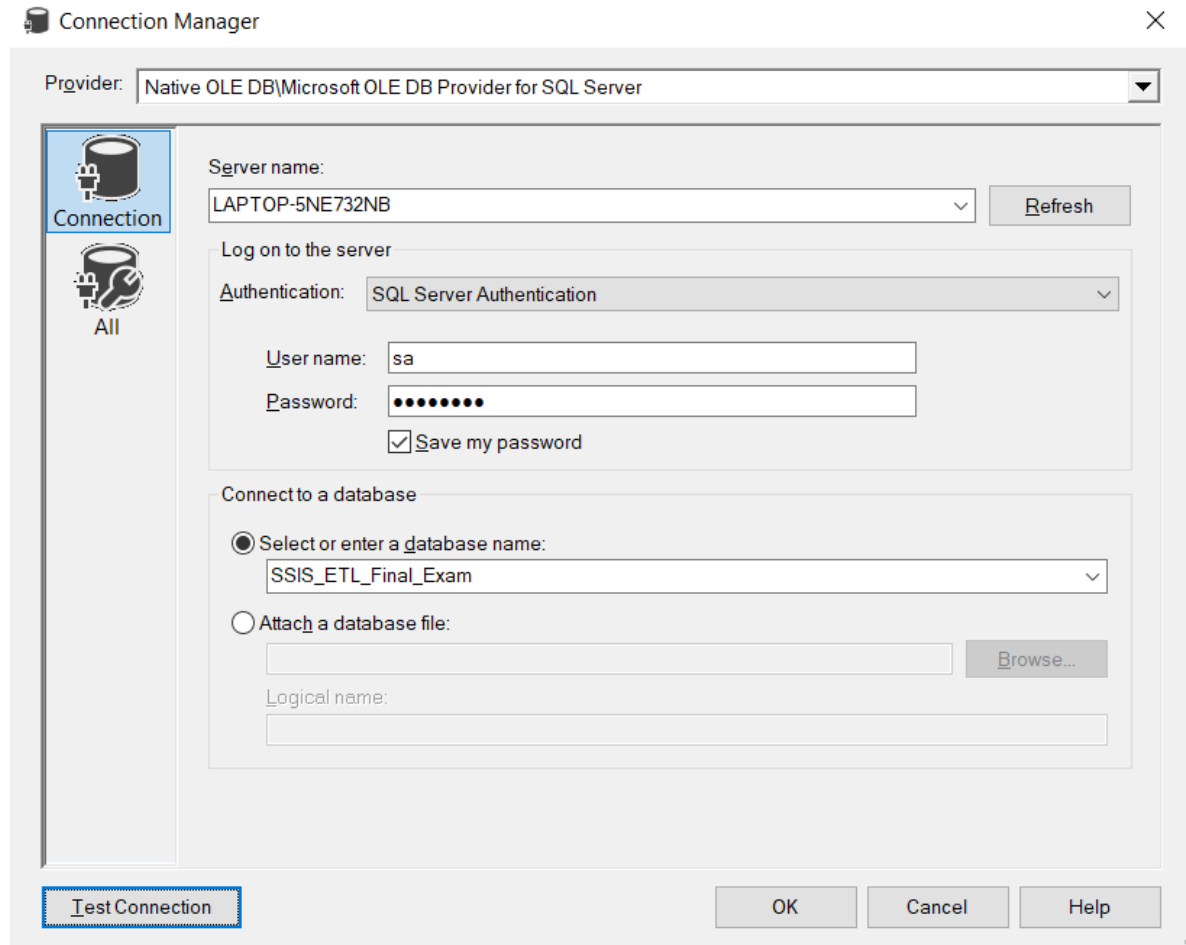


Figure 3.3.1 New Source assistant from Sql Server

3.3.2 New Destination Assistant from Sql Server

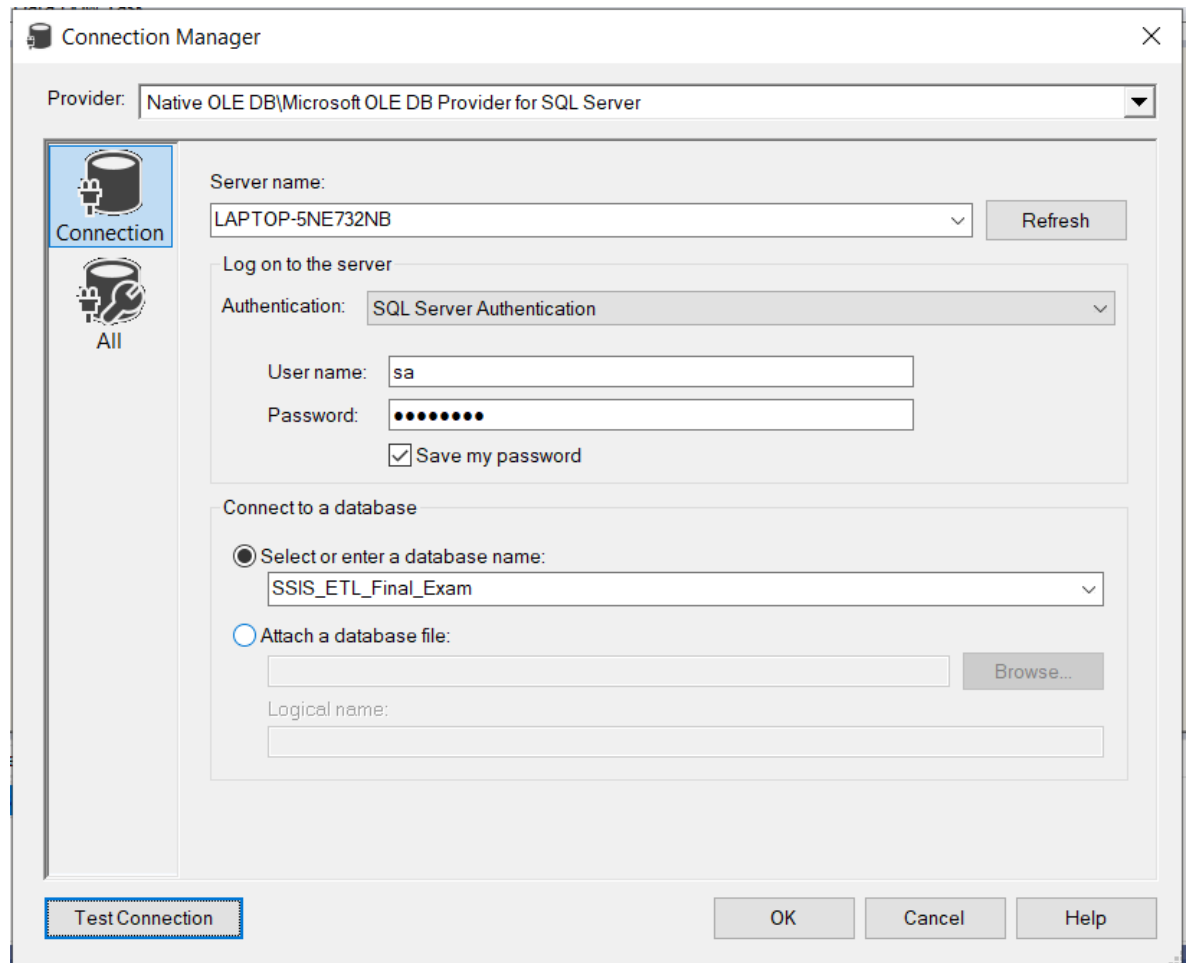


Figure 3.3.2 New Destination assistant from Sql Server

3.3.3 Create data flow for Dimension table

3.3.3.1 DIM_Time

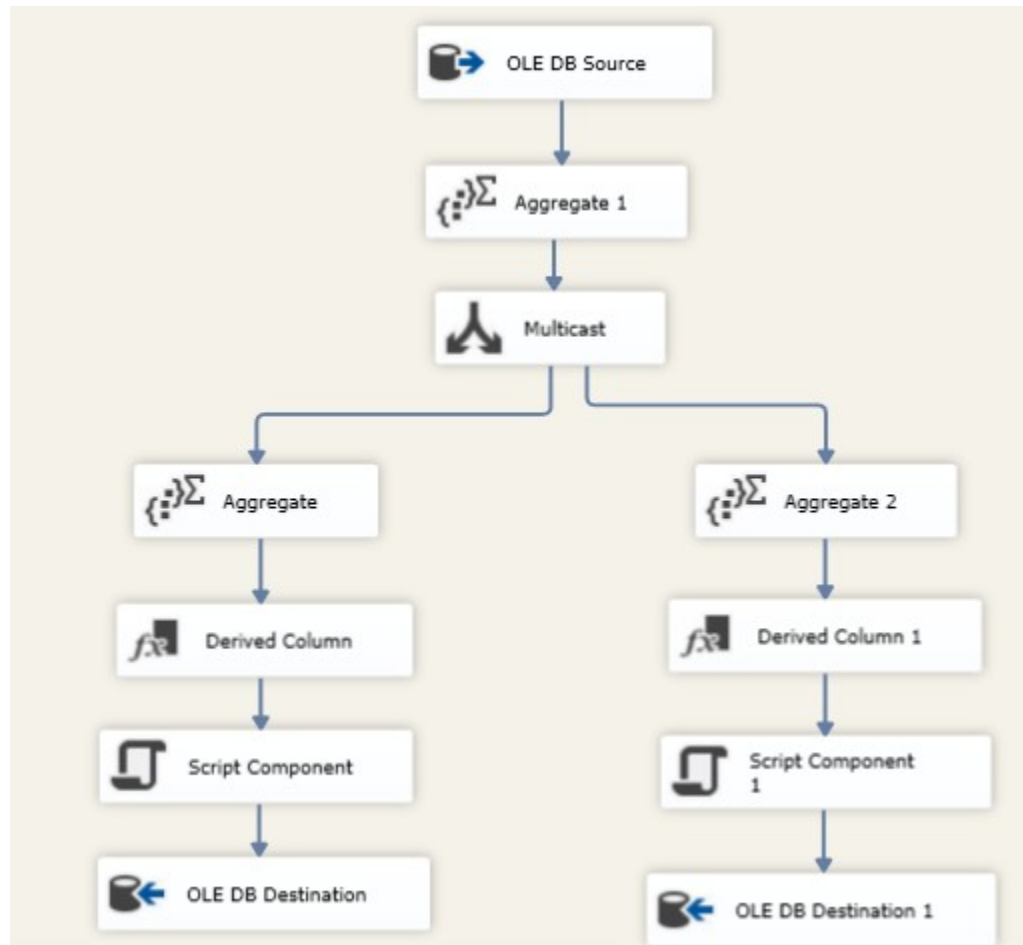


Figure 3.3.3.1 Data flow of DIM_Time

3.3.3.2 DIM_Category

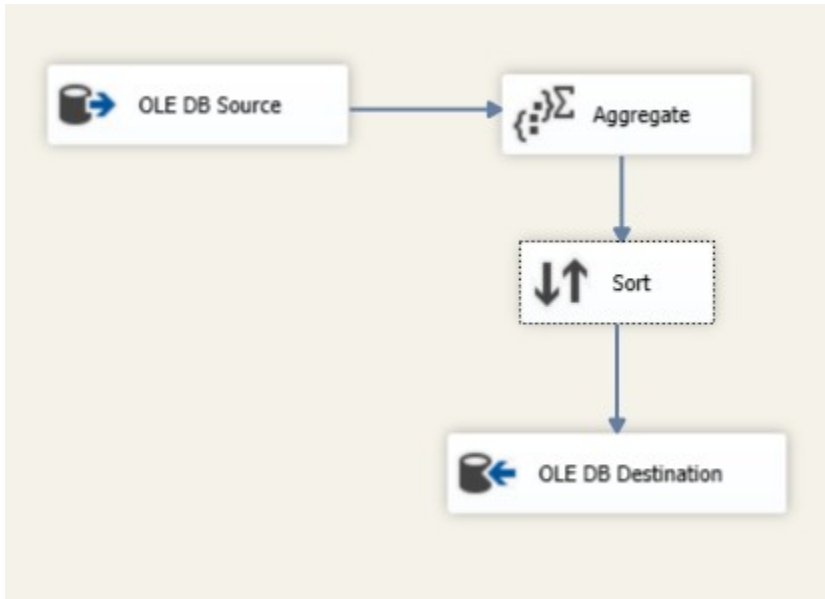


Figure 3.3.3.2 Data flow of DIM_Category

3.3.3.3 DIM_Product

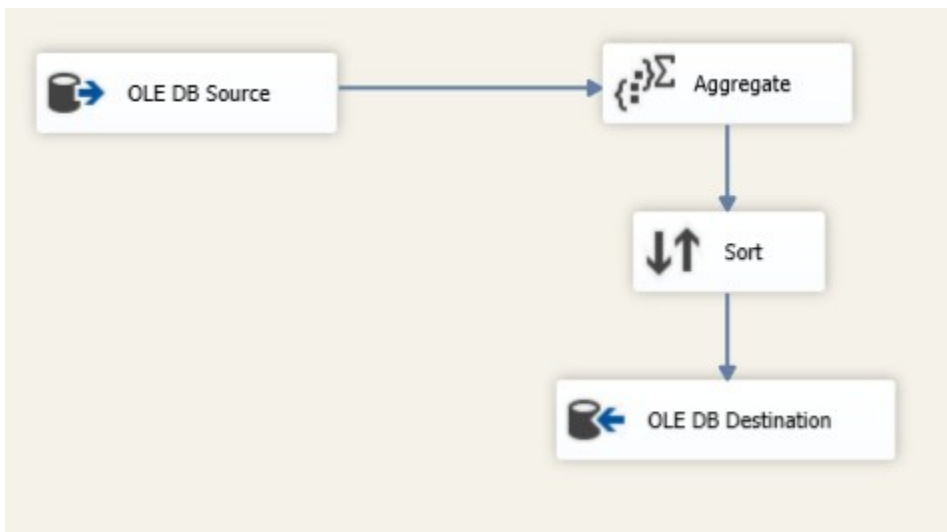


Figure 3.3.3.3 Data flow of DIM_Product

3.3.3.4 DIM_Location

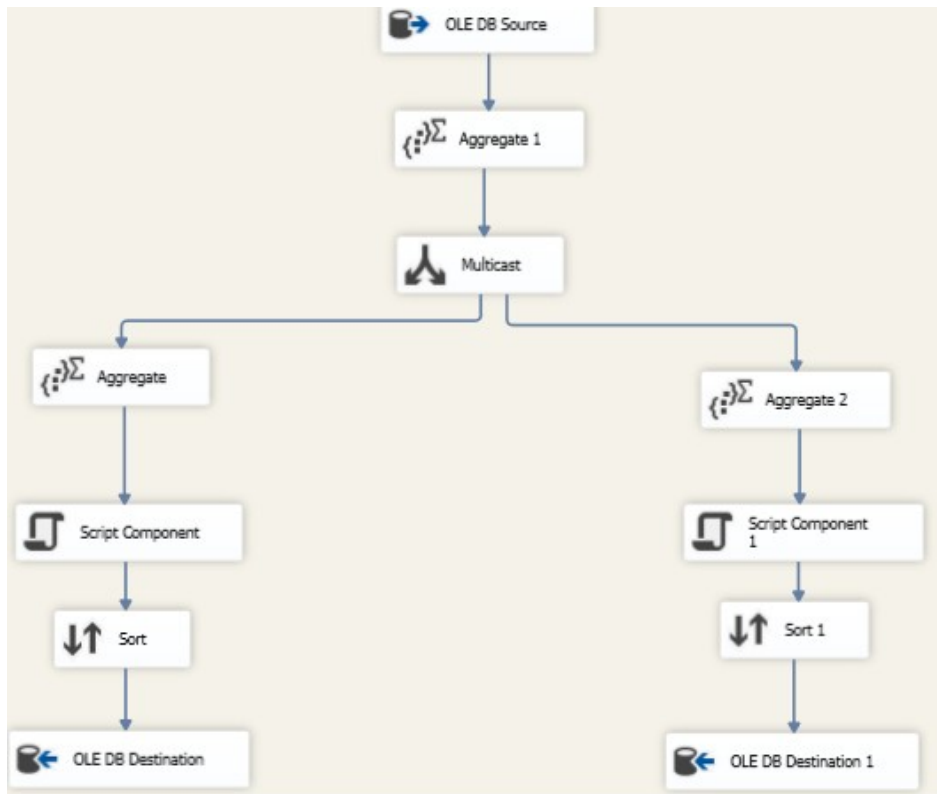


Figure 3.3.3.4 Data flow of DIM_Location

3.3.3.5 DIM_Customer

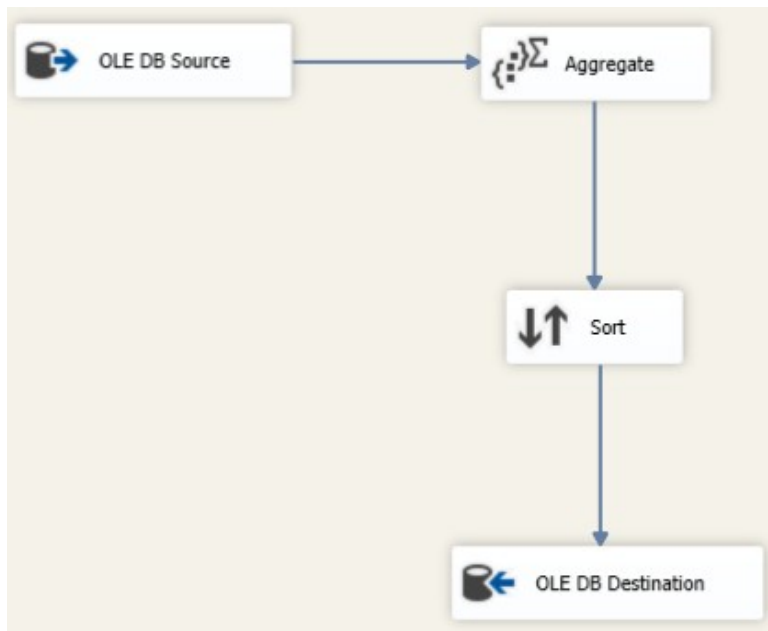


Figure 3.3.3.5 Data flow of DIM_Customer

3.3.4 Create data flow for Fact table

3.3.4.1 FACT_Order

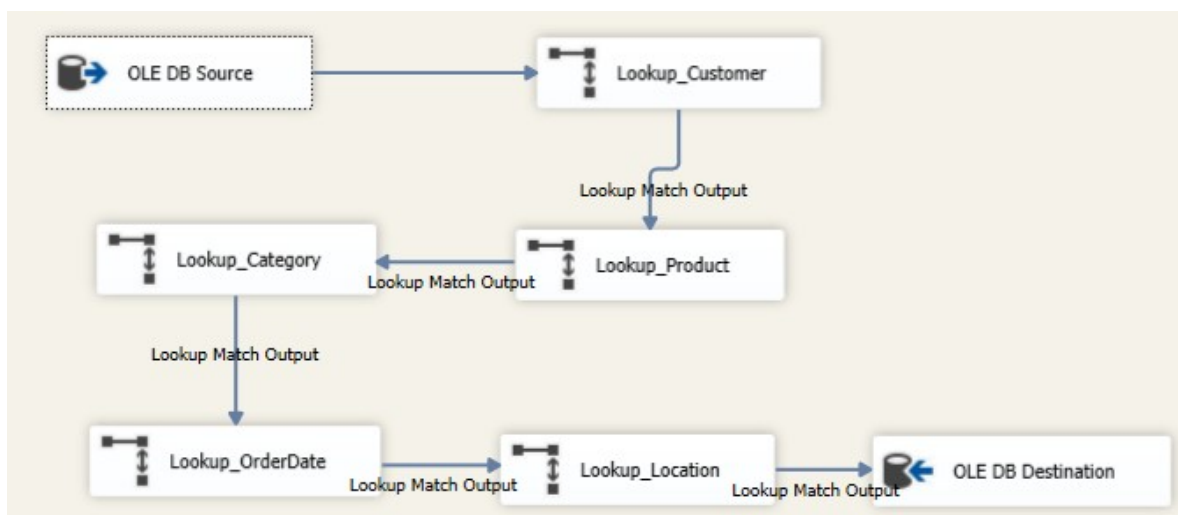


Figure 3.3.4.1 Data flow of FACT_Order

3.3.4.2 FACT_Shipping

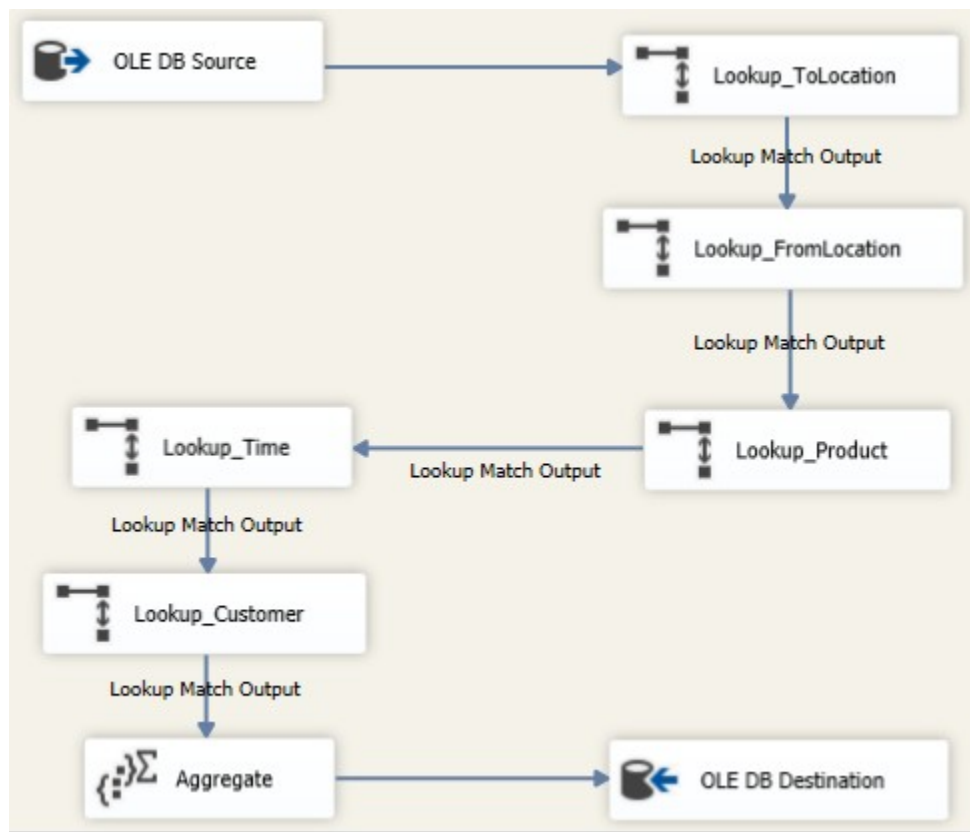


Figure 3.3.4.2 Data flow of FACT_Shipping

3.3.5 Execute SQL Task

3.3.5.1 Delete Foreign Key

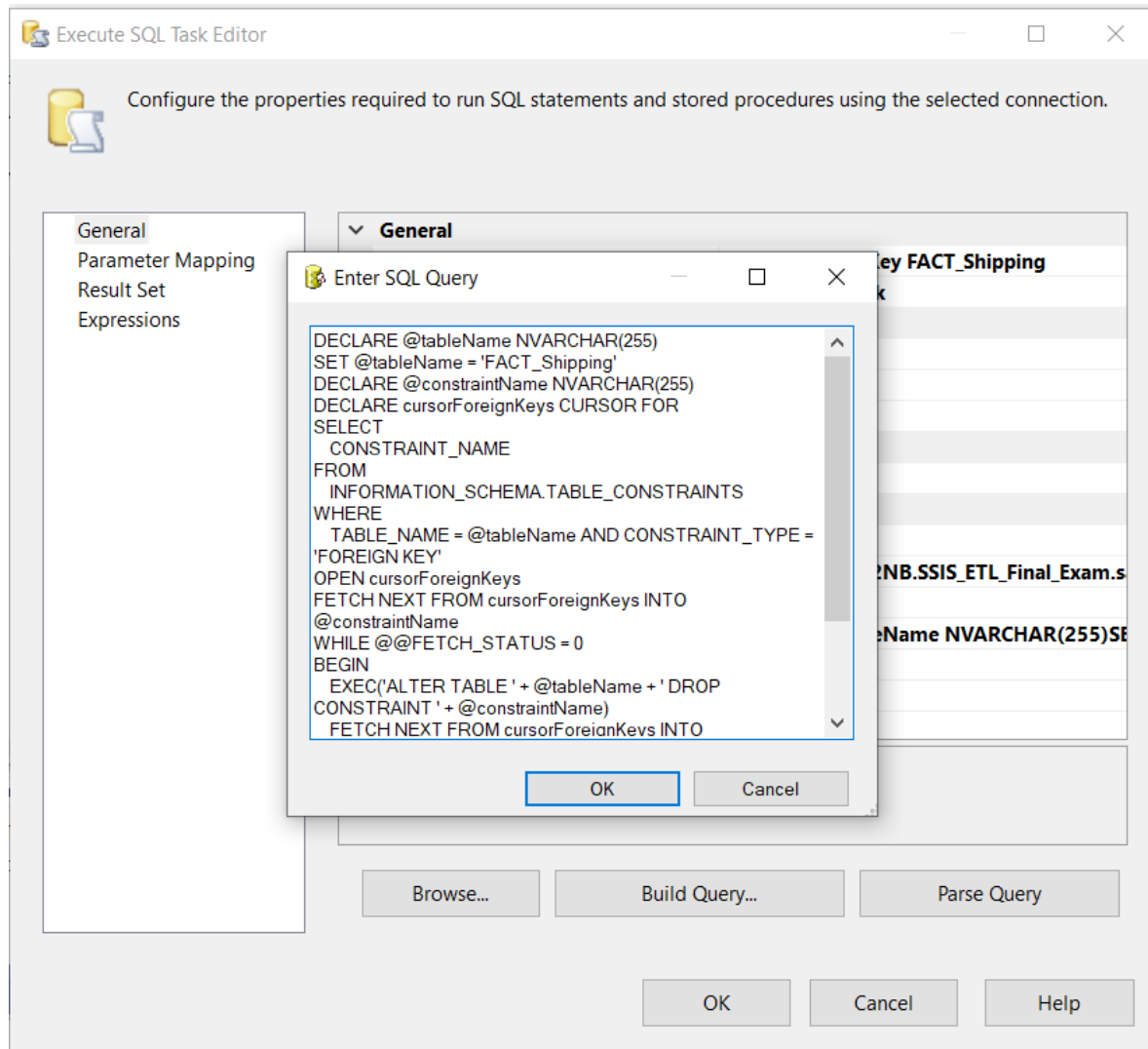


Figure 3.3.5.1 Execute SQL Delete Foreign Key

3.3.5.2 Truncate DIM Table

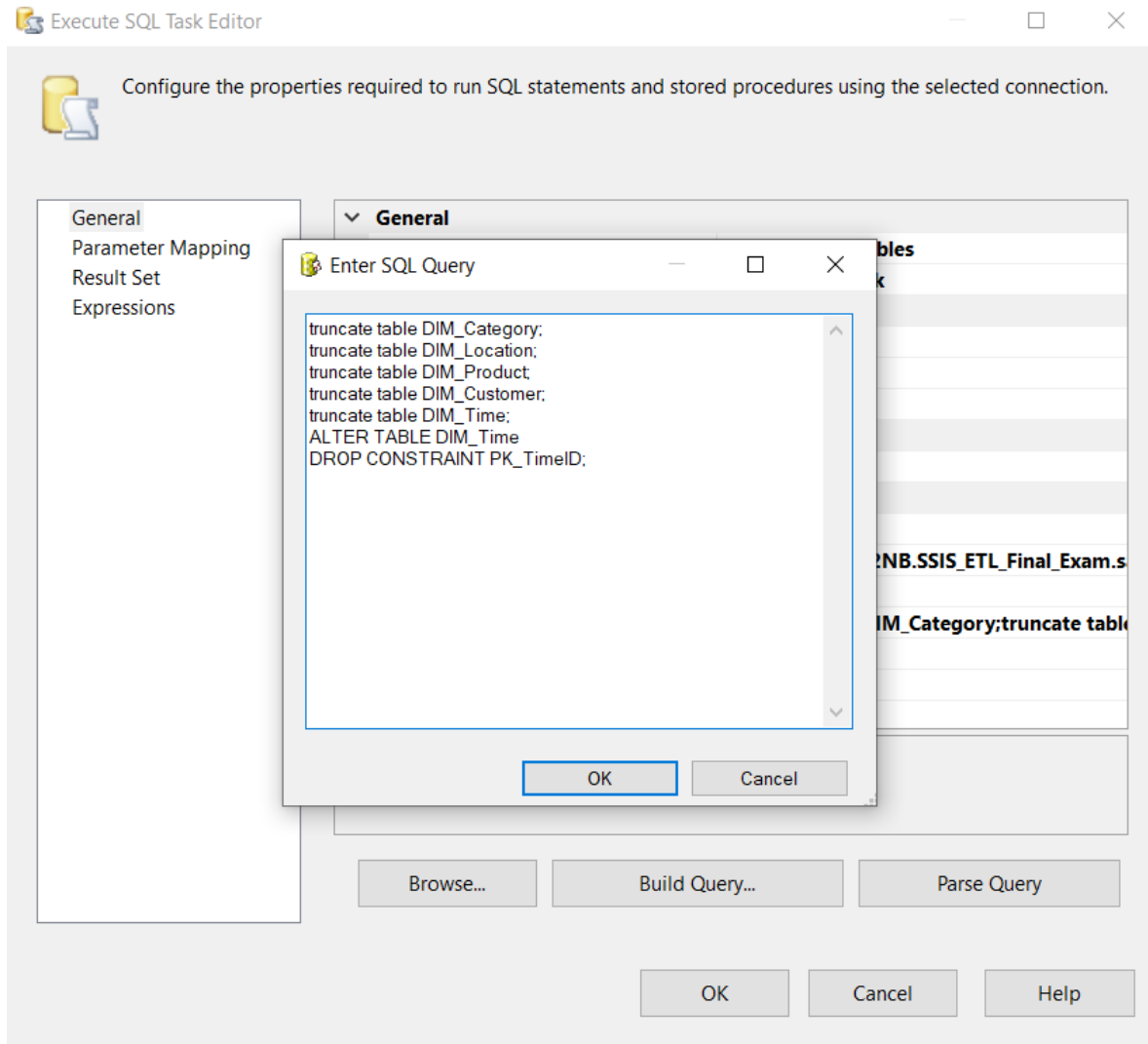


Figure 3.3.5.2 Execute SQL Truncate DIM Table

3.3.5.3 Truncate FACT Table

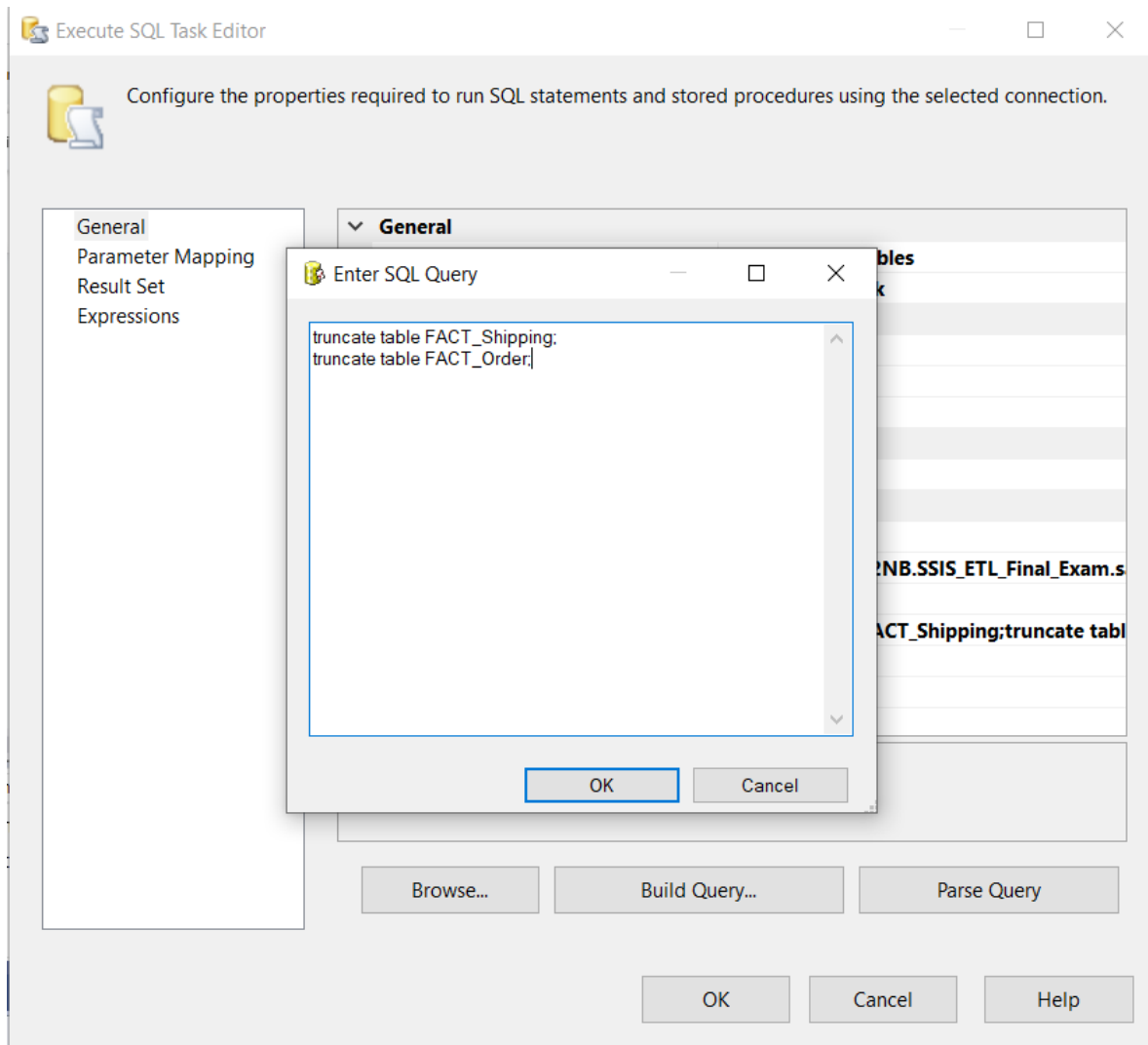


Figure 3.3.5.3 Execute SQL Truncate FACT Table

3.3.5.4 Create Foreign Key FACT_Order

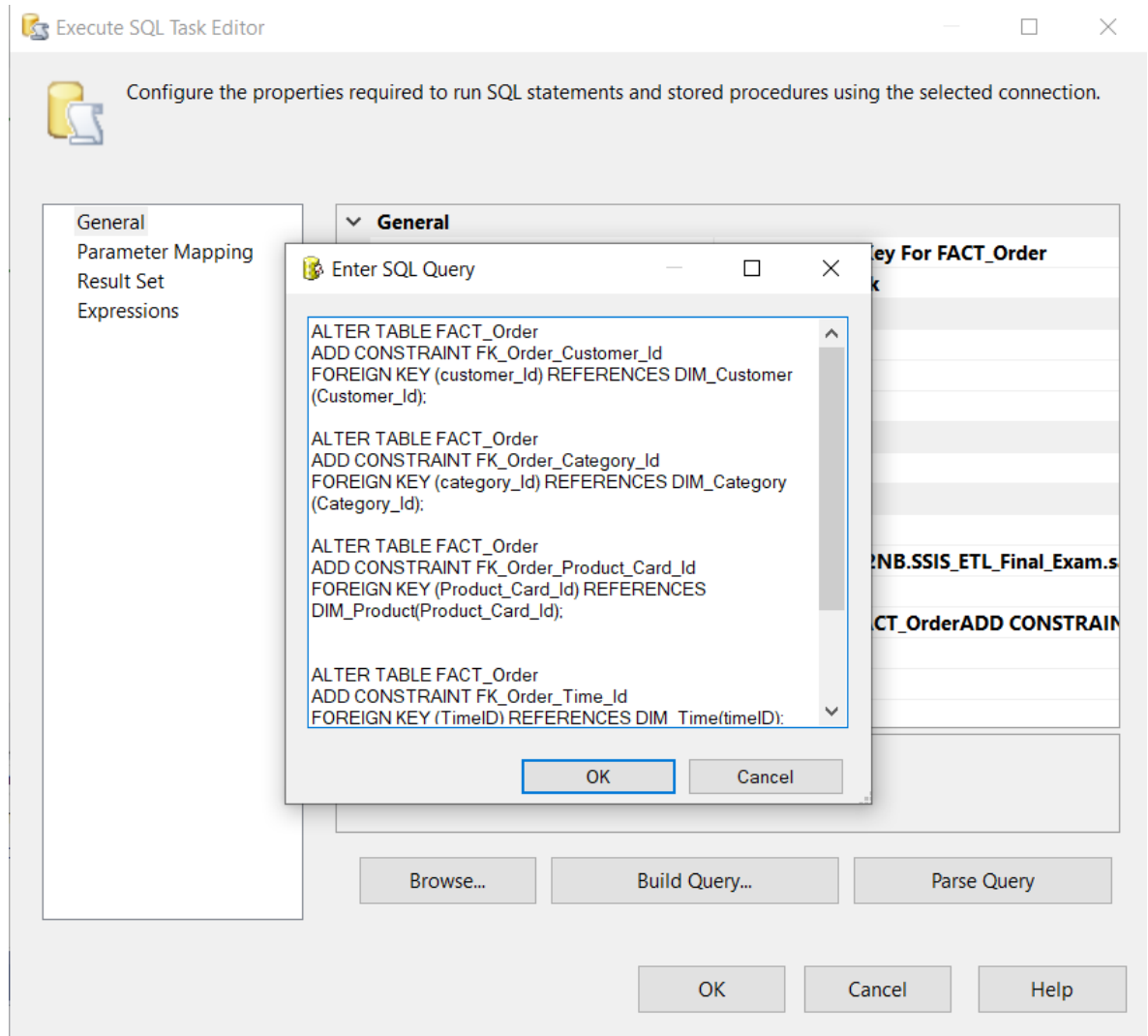


Figure 3.3.5.4 Execute SQL Create Foreign Key FACT_Order

3.3.5.5 Create Foreign Key FACT_Shipping

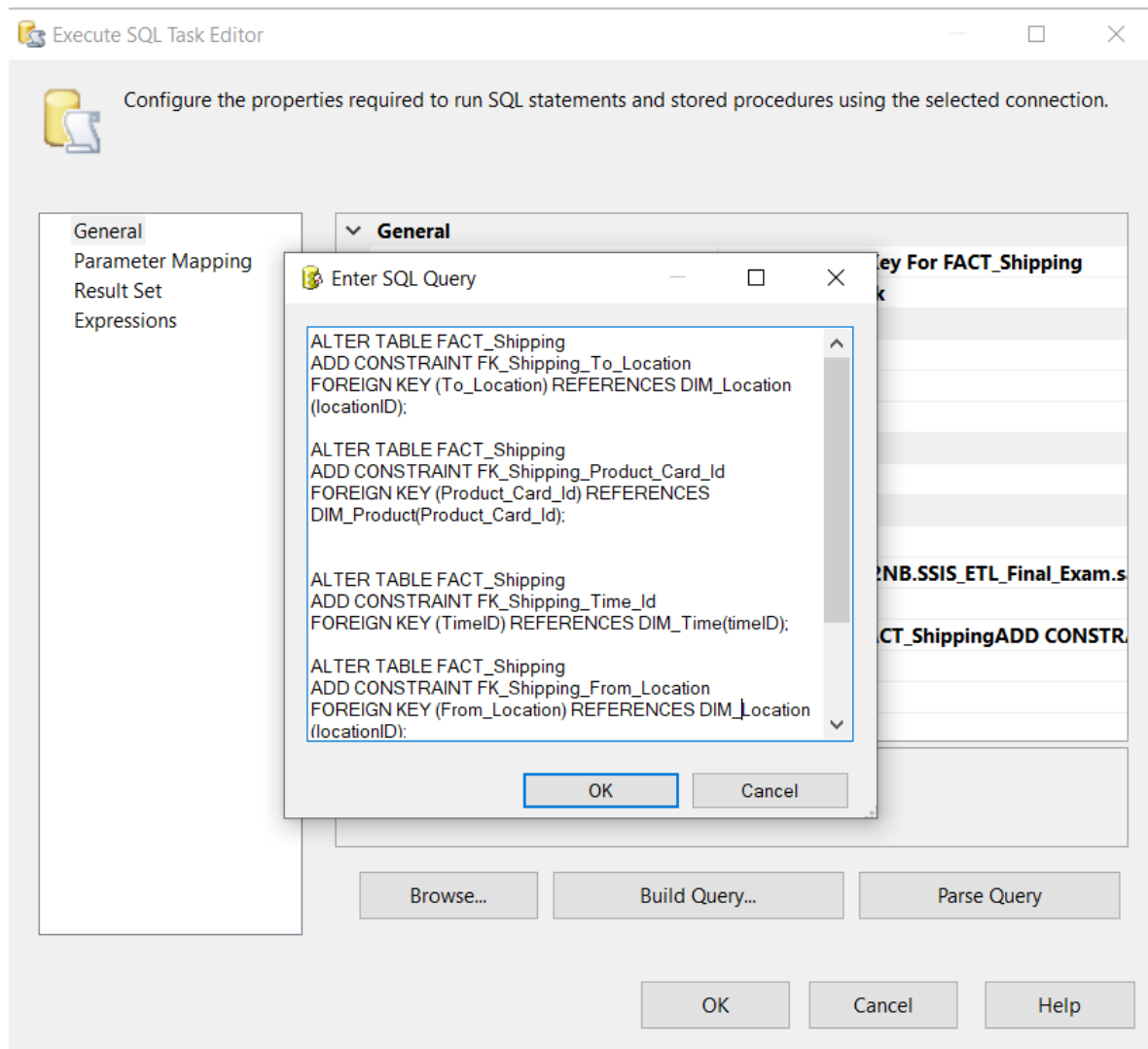


Figure 3.3.5.5 Execute SQL Create Foreign Key FACT_Shipping

3.3.6 Results of running the project

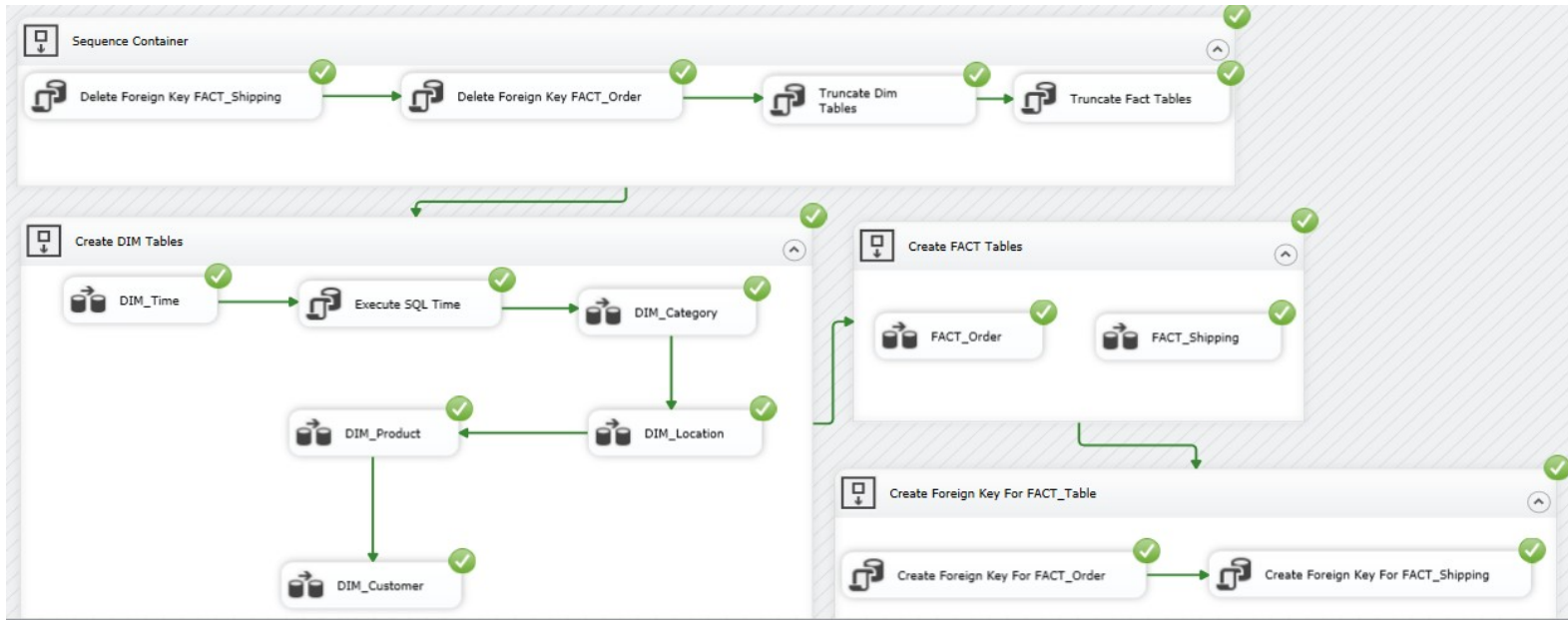


Figure 3.3.6 Results of running the project